

## – Supplementary Material –

### A. Datasets

We conduct experiments on BEAT2 and SHOW. BEAT2 is used for training and in-domain evaluation, whereas SHOW is reserved for zero-shot cross-dataset evaluation without target-domain training or fine-tuning. For methods requiring textual input, transcripts and word-level timestamps are automatically obtained from SHOW audio using a frozen speech-to-text pipeline. The same transcripts and timestamps are supplied to these methods. This protocol evaluates in-domain generation and transfer across datasets with different speakers, motion patterns, and recording conditions.

1) *BEAT2*. BEAT2 [1] provides aligned speech, text, and holistic motion represented with SMPL-X, including facial expressions, body movements, and hand articulation. We use the English BEAT2-Standard subset and follow EMAGE’s 85%/7.5%/7.5% training/validation/test split. Motion is sampled at 30 FPS and represented by 330-dimensional 6D joint rotations, 3D root translation, four foot-contact indicators, and 100-dimensional facial expression coefficients. Audio is sampled at 16 kHz. Training clips contain 64 frames, extracted with a 20-frame stride; the first four frames provide seed motion. We report both *one-speaker* and *all-speaker* results. Unless otherwise stated, BEAT2 is also used for all ablation studies and representation analyses.

2) *SHOW*. SHOW [2] is an in-the-wild holistic co-speech motion dataset reconstructed from monocular talking videos using SMPL-X. Compared with BEAT2, it features different speakers, gesture styles, and recording conditions, providing a cross-dataset test of generalization. We use its test split exclusively for zero-shot evaluation, without training or fine-tuning on SHOW. Methods requiring textual input receive the shared audio-derived transcripts and word-level timestamps described above. All evaluated models retain their BEAT2-trained parameters. The following subsection details the pre-processing, test-set selection, and inference protocols.

### B. Zero-Shot Cross-Dataset Evaluation on SHOW

1) *Evaluation split*. We use the SHOW test partition adopted by TalkSHOW/SemTalk, including all sequences satisfying the pre-processing requirements without performance-based selection. All evaluated checkpoints are trained on the BEAT2 English all-speaker split. SHOW data are not used for training, fine-tuning, checkpoint selection, or parameter updates.

2) *Speech transcription*. For text-conditioned methods, SHOW test audio is transcribed once using WhisperX with `faster-whisper-large-v3`, specifying English. Word-level timestamps are obtained through WhisperX’s alignment procedure, without manual correction. The same transcripts and timestamps are supplied to all text-conditioned methods; audio-only methods retain their original inputs.

3) *Zero-shot inference*. For each method using learned speaker embeddings, we average the embeddings of its BEAT2

training speakers to obtain a fixed speaker condition. This model-specific mean embedding is used for every SHOW sequence, without target-speaker adaptation or identity matching. Methods without speaker embeddings retain their original conditioning scheme. For methods requiring motion history, only the first  $N_{\text{seed}}$  ground-truth frames initialize generation. Subsequent windows use previously generated motion, and seed frames are excluded from evaluation.

4) *Motion preprocessing and evaluation*. SHOW motions are converted to the SMPL-X representation and coordinate convention used for BEAT2. All outputs are evaluated using the same preprocessing, temporal alignment, and metric implementations. We report FGD, BC, DIV, MSE, and LVD. SRGR is omitted because SHOW lacks the frame-level semantic relevance annotations required for its computation.

### C. Additional Implementation Details

All experiments are conducted on a single NVIDIA A100 GPU. Following EMAGE [1], motion is sampled at 30 FPS and represented by 330-dimensional 6D SMPL-X joint rotations, 3D root translation, four foot-contact labels, and 100-dimensional facial expression coefficients. Audio is sampled at 16 kHz. Speech conditioning uses onset and amplitude features, 1,024-dimensional HuBERT representations, 300-dimensional fastText word embeddings, and CLIP ViT-B/32 sentence embeddings.

Training clips contain 64 motion frames with a stride of 20, using the first four frames as seed motion. The body backbone has a hidden dimension of 768, and each regional RVQ latent has 256 dimensions. The speech-to-plan predictor is a two-layer bidirectional Transformer encoder with 128-dimensional features, four attention heads, a feed-forward dimension of 512, and dropout of 0.1.

The future motion-plan tokenizer contains two residual MLP blocks and a codebook with  $K = 64$  entries of dimension  $D_p = 128$ . With horizon  $H = 2$  and offset  $o = 1$ , each plan targets the next two motion tokens, corresponding to eight frames (approximately 0.267 s at 30 FPS). The VQ commitment weight is 0.25.

Both training stages use Adam with  $(\beta_1, \beta_2) = (0.5, 0.999)$ , zero weight decay, and gradient-norm clipping at 0.99. In Stage I, the tokenizer is pretrained for 100 epochs with a batch size of 128 and an initial learning rate of  $2 \times 10^{-4}$ . Its encoder, decoder, and codebook are then frozen. In Stage II, the speech planner, event-aware gate, hierarchical FiLM modules, and trainable body-motion generator are jointly optimized for 200 epochs with a batch size of 64 and an effective initial learning rate of  $5 \times 10^{-5}$ .

Plan selection uses straight-through Gumbel-Softmax with temperature 1.0 during training and deterministic argmax at inference. Plan prediction and its auxiliary losses are disabled for the first 10 epochs, then enabled with the auxiliary loss



Fig. 1. Facial-motion comparison on the same speech segment. Red dashed boxes highlight selected differences in head pose and mouth configuration; the green box outlines PlanTalk’s outputs. The facial pathway follows SemTalk and receives no plan modulation, so this comparison evaluates system outputs rather than a direct planning benefit.

weight linearly increased to its full value over the following 10 epochs.

We set  $\lambda_p = 0.2$ ,  $\lambda_a = 0.1$ ,  $\lambda_c = 0.1$ ,  $\lambda_e = 0.5$ ,  $\lambda_{sem} = 1$ , and  $\alpha_g = 1$ . The contrastive temperature is 0.1. Unless otherwise specified, experiments use random seed 43.

#### D. Qualitative Comparison of Facial Motion

Fig. 1 compares selected facial frames generated for the same speech segment. The methods exhibit different head orientations and mouth configurations, with representative differences highlighted by red dashed boxes. PlanTalk produces visible changes in mouth opening across the displayed speech positions while maintaining a relatively consistent head orientation. Some configurations resemble the ground truth, although differences in mouth opening and expression remain. These selected frames illustrate facial appearance and pose variation; temporal smoothness and lip synchronization require inspection of the corresponding videos.

PlanTalk retains SemTalk’s speech-conditioned facial pathway without modulation by the predicted future plan. This comparison therefore illustrates the facial output of the complete system, rather than establishing a direct benefit of future planning for facial animation. Head-orientation differences should also be distinguished from changes in facial expression and mouth articulation.

#### E. Future Window Design: Horizon Length and Temporal Ordering

1) *Horizon Length and Kinematic Grounding*: We set the window offset to  $o = 1$  and use  $H = 2$  by default, spanning eight frames ( $\approx 0.267$  s at 30 FPS). This short horizon is designed to capture local multi-step motion evolution while limiting the prediction ambiguity associated with more distant future motion.

TABLE I  
EFFECT OF FUTURE-MOTION CONTENT AND TEMPORAL ORDERING. ALL VARIANTS SHARE THE SAME DOWNSTREAM ARCHITECTURE AND TRAINING PROTOCOL.

| Representation             | FGD↓         | BC↑          | DIV↑         | SRGR↑        | Event F1↑    |
|----------------------------|--------------|--------------|--------------|--------------|--------------|
| Future Endpoint Block      | 4.168        | 0.803        | 13.68        | 0.433        | 0.731        |
| Endpoint + Energy          | 4.103        | 0.807        | 13.75        | 0.436        | 0.746        |
| Order-Invariant Future Set | 4.027        | 0.812        | 13.84        | 0.438        | 0.758        |
| Ordered Future Sequence    | <b>3.812</b> | <b>0.817</b> | <b>13.95</b> | <b>0.442</b> | <b>0.784</b> |

TABLE II  
SENSITIVITY TO FUTURE HORIZON LENGTH  $H$  ON BEAT2. ALL VARIANTS SHARE IDENTICAL STAGE I AND STAGE II SETTINGS.

| Horizon $H$    | Duration                  | Plan Acc@1↑  | FGD↓         | BC↑          | SRGR↑        | Event F1↑    |
|----------------|---------------------------|--------------|--------------|--------------|--------------|--------------|
| $H = 1$        | 4 frames (0.133 s)        | <b>0.612</b> | 4.156        | 0.807        | 0.435        | 0.742        |
| $H = 2$ (Ours) | <b>8 frames (0.267 s)</b> | 0.536        | <b>3.812</b> | <b>0.817</b> | <b>0.442</b> | <b>0.784</b> |
| $H = 4$        | 16 frames (0.533 s)       | 0.384        | 4.089        | 0.801        | 0.428        | 0.730        |
| $H = 8$        | 32 frames (1.067 s)       | 0.215        | 4.652        | 0.768        | 0.395        | 0.681        |

Table II evaluates  $H \in \{1, 2, 4, 8\}$  under the same training protocol. With  $H = 1$ , the planner achieves higher Acc@1 (0.612) but lower generation performance than the default setting.  $H = 2$  achieves the best synthesis results across the evaluated metrics, with FGD of 3.812, BC of 0.817, SRGR of 0.442, and Event F1 of 0.784. Increasing the horizon to  $H = 4$  and  $H = 8$  further reduces plan-prediction accuracy and degrades generation quality, consistent with increased ambiguity when predicting more distant motion from speech. These results support the use of a short planning horizon and reveal a trade-off between future coverage and speech-side predictability.

2) *Window Content and Temporal Ordering*: We examine future-window content and ordering through four tokenizer variants under identical downstream settings. Let  $\{u_{t+o+j}\}_{j=0}^{H-1}$  denote tokens in the future window. *Future Endpoint Block* uses only the final token:

$$W_t^{\text{end}} = u_{t+o+H-1}. \quad (1)$$

*Endpoint + Energy* additionally appends the mean motion energy  $\bar{E}_t$ :

$$W_t^{\text{end+energy}} = [u_{t+o+H-1}; \bar{E}_t]. \quad (2)$$

*Order-Invariant Future Set* aggregates all window tokens via shared  $\phi$  and symmetric pooling  $\rho$ :

$$h_t^{\text{set}} = \rho \left( \frac{1}{H} \sum_{j=0}^{H-1} \phi(u_{t+o+j}) \right). \quad (3)$$

In contrast, *Ordered Future Sequence* preserves positional ordering via concatenation:

$$h_t^{\text{ord}} = E_m(\text{Vec}(u_{t+o}, \dots, u_{t+o+H-1})). \quad (4)$$

As shown in Table I, adding energy reduces FGD from 4.168 to 4.103. Order-invariant aggregation further improves FGD to 4.027 and SRGR to 0.438, confirming that window-level motion provides richer guidance than static endpoints. Ordered Future Sequence attains the best performance across all metrics (FGD = 3.812, BC = 0.817, DIV = 13.95, SRGR = 0.442, Event F1 = 0.784), supporting the effectiveness of order-sensitive future-window encoding.

TABLE III  
LINEAR PROBING OF ORACLE AND INFERENCE-TIME REPRESENTATIONS.  
ALL REPRESENTATIONS ARE FROZEN, AND EACH ROW USES THE SAME  
PROBING PROTOCOL.

| Representation                   | Cur. Part<br>F1↑ | Fut. Part<br>F1↑ | Fut. Energy<br>$R^2$ ↑ | Temp. Evol.<br>F1↑ |
|----------------------------------|------------------|------------------|------------------------|--------------------|
| Frozen Speech Feature            | 0.575            | 0.629            | 0.415                  | 0.551              |
| Current Motion Token             | <b>0.792</b>     | 0.558            | 0.327                  | 0.486              |
| Speech-Predicted Future Plan     | 0.582            | 0.697            | 0.544                  | 0.651              |
| Oracle Continuous Future Feature | 0.624            | <b>0.781</b>     | <b>0.684</b>           | <b>0.746</b>       |
| Oracle Discrete Future Plan      | 0.603            | <u>0.754</u>     | <u>0.631</u>           | <u>0.718</u>       |

### F. Understanding the Learned Future Plans

We examine whether future-motion attributes remain accessible in the speech-predicted plans used at inference. We compare the frozen speech feature  $s_t$ , current motion token  $u_t$ , oracle continuous future feature  $h_t^m$ , oracle discrete plan  $p_t^m$ , and predicted plan

$$\hat{k}_t = \arg \max_k l_{t,k}, \quad \hat{p}_t = c_{\hat{k}_t}, \quad (5)$$

where  $\hat{p}_t$  is the codebook embedding supplied to the generator without accessing future motion.

We probe current and future body-part activation, future motion energy, and within-window energy evolution. Regional energy is the mean squared joint velocity within the corresponding block or future window. Region  $r$  is active when its energy exceeds the training-set median  $\tau_r = \text{median}_{\text{train}}(e_r)$ . For  $H = 2$ , temporal evolution is labeled increasing, decreasing, or stable using  $\Delta e_t = e_{t+o+1} - e_{t+o}$ , where  $e_t$  denotes body-motion energy in token block  $t$ . We set  $\delta$  to the training-set 33.3rd percentile of  $|\Delta e_t|$  and use  $\pm\delta$  as thresholds. All thresholds are fixed for validation and testing.

Representations and planners are frozen. We retain the utterance-level splits of the main experiments and standardize features using training-set statistics. Classification probes are affine classifiers trained with class-balanced cross-entropy; future energy is predicted using ridge regression. The ridge regularization coefficient is selected on the validation set from  $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ . We report macro-F1 for classification and  $R^2$  for regression, evaluating only positions with  $m_t = 1$ .

Table III shows that Current Motion Token achieves the highest current-part activation F1 (0.792) but lower scores on future attributes. Frozen Speech Feature outperforms it on future-part activation (0.629 vs. 0.558), future energy (0.415 vs. 0.327), and temporal evolution (0.551 vs. 0.486), suggesting that speech features provide useful cues about upcoming motion.

Speech-Predicted Future Plan improves these scores to 0.697, 0.544, and 0.651, respectively: absolute gains of 0.068, 0.129, and 0.100 over Frozen Speech Feature. Future attributes are therefore more accessible to linear probes in the predicted representation. However, the planner processes the complete speech segment, whereas the baseline probe receives  $s_t$ ; the comparison does not isolate representation learning from the benefit of broader temporal context.

Oracle Continuous Future Feature obtains the highest future-oriented scores of 0.781, 0.684, and 0.746. Quantization

reduces them to 0.754, 0.631, and 0.718, showing that the discrete representation retains useful future-motion structure. The predicted plan trails the oracle discrete plan by 0.057, 0.087, and 0.067, respectively, indicating a remaining prediction gap. Since the predicted plan is a deterministic function of the input speech segment, improved probing scores reflect accessibility to linear readouts rather than additional information introduced at inference.

We contextualize plan-index prediction using codebook utilization and frequency baselines. Training-set assignments use 59 of the 64 codes, with perplexity  $\exp(-\sum_k f_k \log(f_k + \epsilon)) = 39.6$ , where  $f_k$  is the training-set code frequency. The frequency baselines achieve Prior@1 of 0.061 and Prior@5 of 0.226 by selecting the most frequent one and five codes, respectively. The planner achieves Acc@1/Acc@5 of 0.536/0.798 and macro-F1 of 0.468. These comparisons use the same frozen codebook; accuracies across independently learned codebooks are not directly comparable.

### G. Temporal Motion Evolution

We visualize mean frame-to-frame joint displacement as a motion-intensity proxy, distinct from the squared-velocity energy used for probing in Sec. F:

$$d_n = \frac{1}{J} \sum_{j=1}^J \|x_{n,j} - x_{n-1,j}\|_2, \quad (6)$$

where  $J$  is the number of selected body joints and  $x_{n,j}$  is the 3D position of joint  $j$  at frame  $n$ . At a fixed frame rate, larger  $d_n$  indicates greater motion intensity.

Fig. 2 shows a representative 10-second utterance with five selected speech landmarks: *opened* (1.33 s), *shocked* (4.05 s), *everyone* (5.64 s), *surprise* (7.65 s), and *today* (9.14 s). The ground-truth profile contains several pronounced peaks separated by lower-intensity intervals. PlanTalk broadly follows this pattern, although its peak near 4 s occurs earlier and several peaks have lower amplitudes than the reference. Thus, its correspondence is closer at the level of coarse intensity evolution than exact peak timing or magnitude.

StreamTalk captures several high-intensity intervals but exhibits shifted peaks, particularly around 4–5 s and 8–10 s. SemTalk shows narrower responses with lower intensity between peaks, whereas GlobalDiff exhibits less pronounced peaks and frequent local fluctuations. In this example, PlanTalk more closely resembles the reference’s sequence of rising and falling motion intensity. The visualization complements the quantitative evaluation but does not independently establish semantic alignment or isolate the contribution of future planning.

### H. User Study Details

To complement the objective evaluation, we conduct a perceptual user study assessing motion quality and speech-motion consistency. Informed by general subjective assessment principles outlined by the International Telecommunication Union (ITU) [3], we recruit 30 participants to evaluate SemTalk, GlobalDiff, StreamTalk, and PlanTalk. Each method

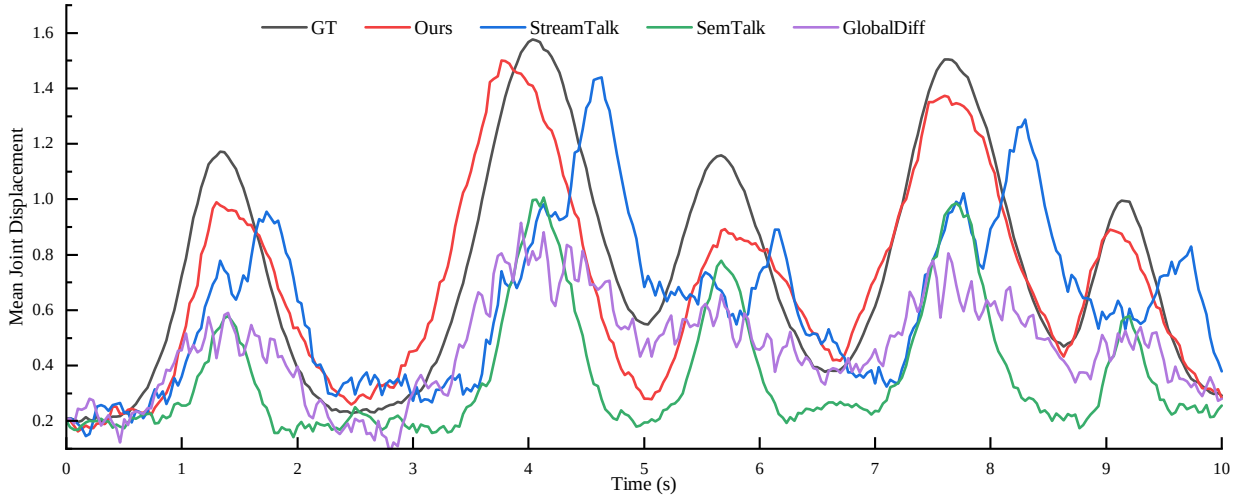


Fig. 2. Mean joint displacement over a representative 10-second utterance. PlanTalk broadly follows the reference intensity profile, while differences in peak timing and amplitude remain.

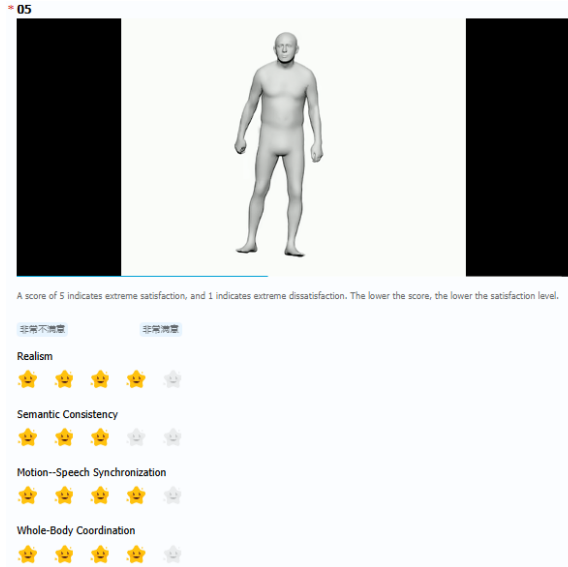


Fig. 3. Screenshot of the user-study interface.

generates motion for the same ten speech sequences, yielding 40 matched videos. Every participant evaluates all videos in randomized order, providing  $30 \times 10 = 300$  ratings per method and perceptual criterion.

Participants rate each video on a five-point scale, with higher scores indicating better perceived quality. Four criteria are considered: *Realism*, assessing motion naturalness and plausibility; *Semantic Consistency*, assessing gesture appropriateness for the speech content; *Motion-Speech Synchronization*, assessing temporal alignment between speech and body motion; and *Whole-Body Coordination*, assessing the coherence of movement across the upper body, hands, torso, and lower body. Participants receive instructions explaining these criteria and the scoring scheme before evaluation. The questionnaire interface is shown in Fig. 3.

For each method and criterion, the mean opinion score (MOS) is computed over all 300 ratings. To construct the error bars, we first average each participant’s ratings over the ten matched sequences, obtaining 30 participant-level mean

scores. Error bars represent the standard deviation across these means, rather than across individual ratings. Since every participant evaluates the same number of sequences, the mean of these participant-level scores equals the MOS computed from all 300 ratings. Inter-rater reliability is assessed using Krippendorff’s alpha, with a reported value of  $\alpha = 0.78$ .

Statistical comparisons use participants, rather than individual ratings, as the unit of analysis. For each criterion, PlanTalk is compared with each competing method using the Wilcoxon signed-rank test on the 30 paired participant-level mean scores. Each pair therefore summarizes the same participant’s judgments of two methods over the same ten speech sequences. This procedure preserves within-participant pairing and avoids treating repeated ratings from one participant as independent observations. The significance threshold is set at  $p < 0.01$ .

## REFERENCES

- [1] H. Liu, Z. Zhu, G. Becherini, Y. Peng, M. Su, Y. Zhou, X. Zhe, N. Iwamoto, B. Zheng, and M. J. Black, “Emage: Towards unified holistic co-speech gesture generation via expressive masked audio gesture modeling,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1144–1154.
- [2] H. Yi, H. Liang, Y. Liu, Q. Cao, Y. Wen, T. Bolkart, D. Tao, and M. J. Black, “Generating holistic 3d human motion from speech,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 469–480.
- [3] International Telecommunication Union, “Methodology for the subjective assessment of the quality of television pictures,” *ITU-R Recommendation BT.500-11*, vol. 4, p. 19, 2002.